

M002 Extension 4 — Complete the Maze

Topic: Add Two AND Conditions · **Course:** Maze Madness · **Level:** Extension (Beginner) · **Time:** about 30 minutes

The agent follows redstone with **four rules**, but gets **stuck** at two spots where two signals are on at once. Add your own **AND** rule there.

1 · The Four Rules

Each rule sees **one** signal and does **one** thing.

```
if redstone on the LEFT:
    turn RIGHT
if redstone on the RIGHT:
    turn LEFT
if redstone BELOW:
    move UP
if redstone ABOVE:
    move DOWN
```

Draw a line to match. You will reuse these.

Signal		Action
redstone left	→	move up
redstone below	→	turn right
redstone above	→	turn left
redstone right	→	move down

2 · AND means BOTH

AND is true only when **both** are true.

```
if redstone LEFT AND redstone BELOW:  
    turn right  
    move up
```

Left on, but NOT below. Is the AND true? Circle one: yes · no

Left on AND below on. Is the AND true? Circle one: yes · no

An AND with two signals needs how many actions? Circle one: one · two · three

3 · Look at the Stuck Spots

Open the **M002 Complete the Maze** world and type `run`. The agent stops at **two** spots. Look around each one.

Tick the signals you see.

	left	right	below	above
Stuck spot 1				
Stuck spot 2				

4 · Write Your Two AND Rules

Use the signals you ticked and the actions from section 1.

Stuck spot 1 — fill the blanks:

```
if redstone _____ AND redstone _____:  
    _____  
    _____
```

Stuck spot 2 — write the whole rule:

5 · Show Your Work 📷 🎥

Type **run**. The goal is to reach the **very end**. If it sticks, fix your AND rules and run again.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.